ADOPD: Answer-Dense On-Policy Distillation under a Reasoning Budget

Method formulation. OPD = on-policy distillation.

1 Motivation

Standard on-policy distillation supervises **every** sampled token with a per-token divergence to a teacher:

$$\mathcal{L}_{\mathrm{OPD}} = \mathbb{E}_{y\sim\pi_\theta}\Big[\textstyle\sum_t \mathrm{KL}\big(\pi_\theta(\cdot \mid y_{<t}) \,\|\, \pi_\varphi(\cdot \mid y_{<t})\big)\Big] \tag{1}$$

This is the right recipe for general capability transfer, but it is actively **counterproductive for efficient reasoning**. If the teacher's trace is longer than the student's budget B, per-token KL on trace positions penalizes the student precisely for compressing. The supervision fights the objective.

The thesis of this method: **the trace is a latent, not a target.**

- The **answer** is what we care about → supervise it densely, per-token, against a strong reference. Low variance, cheap, exactly the quantity of interest.
- The **trace** is a B-token scratchpad whose only job is to make the answer come out right → give it *no* imitation target, let it be telegraphic, notational, or unlike the teacher's prose, and optimize it only through its downstream effect.

This buys two things at once: freedom for the trace to compress into whatever encoding is efficient, and a dense (not sparse-binary) learning signal that is far lower variance than outcome RL.

Rate–distortion framing. The trace is a B-token channel between the model's latent computation and its answer. We minimize answer distortion subject to rate $\leq B$. Sweeping B traces out a rate–distortion curve, which is the natural evaluation object for this project.

2 Setup and notation

Symbol	Meaning
x	prompt
$c = (c_1, \ldots, c_{\lvert c\rvert})$	reasoning trace, $\lvert c\rvert \leq B$ CoT
$a = (a_1, \ldots, a_{\lvert a\rvert})$	answer segment
π_θ	student (trainable)
π_φ	teacher (frozen)
B	hard reasoning-token budget
G	rollouts per prompt (group size)

Student rollout is two-phase and on-policy: $c \sim \pi_\theta^B(\cdot \mid x)$, then $a \sim \pi_\theta(\cdot \mid x, c)$.

Per-token answer divergence (the dense signal):

$$D(x, c, a) \;=\; \frac{1}{|a|} \sum_{t=1}^{|a|} \mathrm{KL}\Big(\pi_\theta(\cdot \mid x, c, a_{<t}) \,\Big\|\, p^*(\cdot \mid x, a_{<t})\Big) \tag{2}$$

p^* is the teacher reference answer distribution (§4). Reverse KL (student ‖ teacher) is deliberate: mode-seeking yields decisive, crisp answers rather than hedged mass-covering ones. Both sides are full distributions at each position, so the KL is computed **exactly** — no sampling estimator needed, given teacher logits.

Objective:

$$\min_\theta \; J(\theta) \;=\; \mathbb{E}_x \, \mathbb{E}_{c \sim \pi_\theta^B(\cdot|x)} \, \mathbb{E}_{a \sim \pi_\theta(\cdot|x,c)} \big[\, D(x, c, a) \,\big] \tag{3}$$

3 The gradient: an asymmetric estimator

This is the technical core. The score-function expansion has three terms:

$$\nabla_\theta J = \underbrace{\mathbb{E}\big[\nabla_\theta D\big]}_{\text{(i) dense, pathwise}} + \underbrace{\mathbb{E}\big[D \cdot \nabla_\theta \log \pi_\theta(c \mid x)\big]}_{\text{(ii) trace visitation}} + \underbrace{\mathbb{E}\big[D \cdot \nabla_\theta \log \pi_\theta(a \mid x, c)\big]}_{\text{(iii) answer visitation}} \tag{4}$$

We keep (i) and (ii), and drop (iii). The asymmetry is the method.

- **Drop (iii).** Every answer position already carries its own pointwise target; the per-position objective is minimized pointwise and the visitation term merely reweights states. This is the standard $\gamma = 0$ / no-credit-assignment choice in GKD and OPD. Keeping it adds variance for no signal.
- **Keep (ii).** Trace positions have *no* pointwise target. The visitation term is not a variance-reduction nicety here — **it is the entire learning signal for the trace.** Drop it and the reasoning policy never updates. Any implementation of "dense supervision only on the answer" that omits this term silently trains nothing but the answer head.

So the method uses **mixed-horizon credit assignment**: $\gamma = 0$ within the answer (dense per-token targets), $\gamma = 1$ across the trace (one terminal reward for the whole scratchpad). One sentence, whole algorithm.

Two channels reach the trace. Term (i) backpropagates into trace parameters through the transformer context (the trace's keys/values are in scope at answer positions), shaping *how a given trace is encoded*. Term (ii) shapes *which trace gets sampled*. Both are necessary; neither substitutes for the other.

Variance reduction. Sample G traces per prompt and use a leave-one-out / group baseline (RLOO / GRPO-style):

$$A^{(g)} = -\frac{\big(\tilde{D}^{(g)} - \mu\big)}{\sigma}, \qquad \tilde{D}^{(g)} = D^{(g)} + \lambda \frac{|c^{(g)}|}{B} \tag{5}$$

with μ, σ the group mean/std, and A stop-gradiented. The sign is such that **lower divergence ⇒ positive advantage**. The λ term makes B a ceiling rather than a target, applying pressure to finish early.

Surrogate loss (what you actually implement):

$$\mathcal{L}(\theta) = \frac{1}{G}\sum_{g}\Big[\underbrace{D^{(g)}}_{\text{answer, differentiable}} - \underbrace{A^{(g)} \cdot \frac{1}{|c^{(g)}|}\sum_{t\in c^{(g)}} \log \pi_\theta\big(c_t^{(g)} \mid \cdot\big)}_{\text{trace, REINFORCE}}\Big] \tag{6}$$

Add PPO-style ratio clipping on the trace term if taking multiple optimizer epochs per batch. Optionally add $\beta\,\mathrm{KL}(\pi_\theta \| \pi_{\text{ref}})$ on trace positions to guard legibility (§6).

4 Choosing the teacher reference p^*

Two variants, differing in what the teacher conditions on. This choice determines what the trace is being optimized *for*.

here a is from the student model.

V2 — Reference teacher (recommended default). $p^*(\cdot \mid x, a_{<t}) = \pi_\varphi(\cdot \mid x, c^*, a_{<t})$ where c^* is the teacher's own **full-budget** trace, sampled once per prompt offline and cached.

> Target semantics: *"match the answer distribution you would have had if you had thought long."* Only the student side is budget-constrained, so all compression pressure lands on the student's trace. This is self-distillation of long-CoT into short-CoT when π_φ is the same model at large budget — the cleanest framing for an efficiency paper.

V1 — Reader teacher. $p^*(\cdot \mid x, a_{<t}) = \pi_\varphi(\cdot \mid x, c, a_{<t})$ — teacher reads the *student's* trace.

> Target semantics: *"does this trace carry enough information for a strong reader?"* Better for trace informativeness and interpretability. Weaker for pure efficiency, and exposed to the misleading-trace failure (§6).

Interpolating (p^* a mixture, or D a convex combination) is reasonable; V2 for the compression signal, V1 for trace-faithfulness pressure.

Compute note — this is cheaper than vanilla OPD. Vanilla OPD needs teacher logprobs over the entire long trace. Here the teacher scores only $|a|$ positions. In V2, c^* is cached and its KV reused across all G rollouts of a prompt and across epochs, so marginal teacher cost per rollout $\approx |a|$ tokens. Dense supervision is concentrated exactly where it is cheap.

5 Budget enforcement

Mechanism — hard truncation with a halt token. Generate c freely; the model may emit `</think>` on its own. At $|c| = B$, force-inject `</think>` and transition to the answer. Compute is exactly bounded, and traces clipped mid-thought produce a bad answer → low reward → the policy learns to halt gracefully before the ceiling. Mask force-injected tokens from the REINFORCE logprob sum (they were not policy choices) while still letting the resulting D penalize that trace.

Budget-conditional training (recommended). Prepend a budget tag (`<budget:512>`) and sample $B \sim \{128, 256, 512, 1024, 2048\}$ per rollout. Yields a single model with a test-time compute dial, shares data across budgets, and makes the rate–distortion curve a property of one checkpoint rather than a sweep of runs.

Annealing. Early in training, aggressive budgets are catastrophic and the signal degenerates. Start the budget distribution near the teacher's typical trace length, anneal the mass toward B_{target}.

6 Failure modes and mitigations

Failure	Mechanism	Mitigation
Agreement on error (V1)	A misleading trace steers teacher and student to the same wrong answer → low D, high reward	Use V2; or gate with verifiable reward: $A \mathrel{+}= \beta \mathbb{1}[\text{correct}]$ on prompts with ground truth
Trace no-op collapse	If the teacher can answer without reasoning, D is $\sim$invariant to c; λ then drives $\lvert c\rvert \to 0$	Correct behavior on easy prompts (adaptive compute). Pathological only if teacher $\gg$ student — keep teacher capability comparable, or use V2 where the student side is the constrained one
Illegible trace drift	No imitation target ⇒ trace may collapse to compressed non-language	Often *desirable* for pure efficiency. If legibility is required, add $\beta\,\mathrm{KL}(\pi_\theta \Vert \pi_{\text{ref}})$ on trace tokens. Treat β as an explicit legibility-vs-efficiency knob, and report the tradeoff
Length bias in D	Sum-over-tokens lets long answers dominate	Mean over answer tokens (as defined); report sensitivity
Teacher log-prob access	API teachers expose only top-k logprobs	Top-k truncated KL with a tail bucket, or a sampled estimator. Prefer a local teacher
Reward hacking of λ	Emitting `</think>` immediately maximizes the length term	Normalize λ against the D scale; the group-std normalization already helps; sanity-check mean trace length per step

7 Algorithm

Algorithm 1 ADOPD training loop

Require: student π_θ, teacher π_φ, prompts $\mathcal{D}$, budget schedule B_k, group size G, length coef λ, legibility coef β, clip ε

Offline (V2): for each $x \in \mathcal{D}$: sample $c^* \sim \pi_\varphi(\cdot \mid x)$ at full budget; cache c^* and its KV
for step $k = 1, 2, \ldots$ **do**
  sample prompt batch $X \subset \mathcal{D}$
  for each $x \in X$ **do**
    sample budget $B \sim \text{Schedule}(k)$ ▷ budget-conditional
    for $g = 1, \ldots, G$ **do**
      // rollout (no grad)
      $c^g \leftarrow$ generate from $\pi_\theta(\cdot \mid x, B)$ until `</think>` or $|c| = B$
      if truncated **then** force-append `</think>`, mark position
      end if
      $a^g \leftarrow$ generate from $\pi_\theta(\cdot \mid x, c^g)$ until EOS
      // teacher scoring: $|a|$ positions only
      $P^* \leftarrow \pi_\varphi$ logits over answer positions of a^g
        given $(x, c^*, \cdot)$ [V2] or $(x, c^g, \cdot)$ [V1]
    end for
    // student forward, with grad
    for $g = 1, \ldots, G$ **do**
      $Q \leftarrow \pi_\theta$ logits over trace and answer positions of (c^g, a^g)
      $D^g \leftarrow \text{mean}_t \, \text{KL}(Q_t \,\|\, P_t^*)$ over answer positions
      $\tilde{D}^g \leftarrow D^g + \lambda \cdot |c^g| / B$
    end for
    // group-relative advantage for the trace
    $\mu, \sigma \leftarrow$ mean, std over $\{\tilde{D}^g\}$
    $A^g \leftarrow -(\tilde{D}^g - \mu)/\sigma$ ▷ stop-grad
    // loss
    $L \leftarrow \text{mean}_g \big[D^g - A^g \cdot \text{mean}_{t \in c^g, \, \text{unmasked}} \log \pi_\theta(c_t^g \mid \cdot) + \beta \cdot \text{KL}(\pi_\theta \| \pi_{\text{ref}}) \text{ on trace positions} \big]$
  end for
  $\theta \leftarrow \theta - \eta \nabla_\theta L$ ▷ PPO clipping on the trace term if multi-epoch
end for

Implementation sketch

```
# q_logits: [G, T, V] student, full sequence (trace + answer)
# p_logits: [G, Ta, V] teacher, answer positions only
# ans_mask: [G, T] bool, True on answer positions
# trc_mask: [G, T] bool, True on trace positions the policy actually chose

logq = q_logits[ans_mask].log_softmax(-1)
logp = p_logits.reshape(-1, V).log_softmax(-1)
kl_tok = (logq.exp() * (logq - logp)).sum(-1) # exact reverse KL
D = scatter_mean(kl_tok, seq_idx, dim_size=G) # [G] mean over answer tokens

D_til = D + lam * trace_len / B
```

```
A = -((D_til - D_til.mean()) / (D_til.std() + 1e-6)).detach()

trace_lp = gather_logprobs(q_logits, tokens)[trc_mask]
trace_lp = scatter_mean(trace_lp, seq_idx, dim_size=G) # [G]

loss = (D - A * trace_lp).mean() + beta * trace_kl_to_ref
```

The only subtle part is that `D` keeps its graph (pathwise, term i) while `A` is detached (score-function, term ii). Getting that backwards is the one bug that will quietly produce a model that never learns to reason shorter.

8 Evaluation plan

Primary object: the rate–distortion curve — accuracy vs. B on AIME / MATH-500 / GPQA-Diamond. Summarize with area-under-curve and *tokens-to-reach-X%*.

Baselines

1. Vanilla OPD (dense on all tokens) + truncation at inference — the direct comparison the thesis predicts we beat.
2. SFT on teacher traces compressed by an LM summarizer.
3. Outcome RL (GRPO) with a length penalty — the sparse-reward comparison.
4. Inference-only budget forcing (s1-style).
5. Student at full budget / teacher at full budget — floor and ceiling.

Metrics accuracy@B; answer-KL to reference; budget adherence (fraction halting unforced); overthinking rate (tokens spent past the point the answer stops changing); trace legibility (LM judge).

Ablations — in priority order

1. **Remove the trace REINFORCE term (ii).** Tests the central claim of §3. Prediction: trace length and structure barely move; only the answer head adapts.
2. V1 vs. V2 teacher.
3. λ sweep (length pressure) and β sweep (legibility pressure) — report the efficiency/legibility frontier.
4. Budget-conditional vs. fixed-B training.
5. Annealing schedule vs. constant budget.
6. Reverse vs. forward KL on the answer.
7. Answer-dense KL vs. sparse binary correctness as the trace reward (isolates the variance benefit of density).

9 Relation to existing work

- **On-policy distillation / GKD** — same on-policy, dense-per-token skeleton; ADOPD restricts density to the answer span and reintroduces the visitation term for the trace.
- **Latent-variable / variational reasoning** — c is a latent; term (ii) is exactly the score-function estimator for it. The objective is an expected-distortion (not ELBO) form.
- **Outcome RL for reasoning (GRPO et al.)** — same group-baseline machinery, but the reward is a dense continuous divergence rather than a binary verifier, which is the variance argument.
- **Efficient-reasoning / budget-forcing** — those act at inference; ADOPD trains the budget in, and makes it steerable via budget conditioning.
- **Information bottleneck / rate–distortion** — B is rate, D is distortion; §7's primary metric is literally the R–D curve.

10 Open questions

1. Does the trace, freed from imitation, drift to a compressed non-linguistic encoding — and does that encoding *transfer* across tasks or overfit the training distribution?
2. Is D (answer-distribution KL) a better trace reward than binary correctness even when ground truth is available? The variance argument says yes; the reward-hacking argument says watch for teacher-agreement on errors.
3. Under budget conditioning, does a single model reach the R–D frontier of the per-B specialist models, or is there a conditioning tax?
4. Does the mixed-horizon estimator ($\gamma = 0$ answer / $\gamma = 1$ trace) beat a uniform-γ treatment, or is there an intermediate discount on the trace that works better?
5. Self-distillation limit: how far can B compress before the student's short-trace answer distribution provably cannot match the long-trace reference (an irreducible-distortion floor)?

A Derivation of the gradient decomposition

This appendix derives the three-term split used in §3 from first principles.

A.1 Objective as an expectation over the joint rollout

Write $J(\theta)$ as an expectation over the joint sample $z = (c, a)$:

$$J(\theta) = \iint \pi_\theta(c \mid x)\, \pi_\theta(a \mid x, c)\, D(x, c, a)\, dc\, da \tag{7}$$

Let $p_\theta(z) = \pi_\theta(c \mid x)\pi_\theta(a \mid x, c)$; this factorization is just the chain rule for the autoregressive rollout (sample the trace, then sample the answer conditioned on it, both from the same policy). Crucially, the integrand $D(x, c, a) = D_\theta(x, c, a)$ also depends on θ: by definition

$$D(x, c, a) = \frac{1}{|a|} \sum_{t=1}^{|a|} \mathrm{KL}\Big(\pi_\theta(\cdot \mid x, c, a_{<t}) \,\Big\|\, p^*(\cdot \mid x, a_{<t})\Big) \tag{8}$$

has π_θ appearing directly inside it. So $J(\theta) = \mathbb{E}_{z\sim p_\theta}[f_\theta(z)]$ with $f_\theta(z) = D(x, c, a)$, and *both* the sampling distribution and the integrand carry θ-dependence.

A.2 General identity for $\nabla_\theta \mathbb{E}_{z\sim p_\theta}[f_\theta(z)]$

Differentiate under the integral sign and apply the product rule:

$$\nabla_\theta \int p_\theta(z) f_\theta(z)\, dz = \int \big[\nabla_\theta p_\theta(z)\big] f_\theta(z)\, dz \;+\; \int p_\theta(z)\, \big[\nabla_\theta f_\theta(z)\big]\, dz \tag{9}$$

The second term is already an expectation under p_θ of $\nabla_\theta f_\theta(z)$ — differentiating f_θ while treating the sampled z as fixed data. The first term uses the log-derivative (score-function) trick,

$$\nabla_\theta p_\theta(z) = p_\theta(z)\, \nabla_\theta \log p_\theta(z), \tag{10}$$

which turns it back into an expectation under p_θ as well:

$$\nabla_\theta J(\theta) = \mathbb{E}_{z\sim p_\theta}\big[f_\theta(z)\, \nabla_\theta \log p_\theta(z)\big] \;+\; \mathbb{E}_{z\sim p_\theta}\big[\nabla_\theta f_\theta(z)\big] \tag{11}$$

This is the general split between "how likely was this sample" (score-function term) and "what does the integrand compute" (pathwise term). No approximation has been made yet — Eq. (11) is exact.

A.3 Splitting the score-function term via the rollout factorization

Because $p_\theta(z) = p_\theta(c, a)$ factorizes autoregressively,

$$\log p_\theta(c, a) = \log \pi_\theta(c \mid x) + \log \pi_\theta(a \mid x, c) \tag{12}$$

so its gradient splits additively:

$$\nabla_\theta \log p_\theta(z) = \nabla_\theta \log \pi_\theta(c \mid x) \;+\; \nabla_\theta \log \pi_\theta(a \mid x, c) \tag{13}$$

Substituting into Eq. (11) and relabeling $f_\theta(z) = D(x, c, a)$ gives the three-term decomposition used in §3:

$$\nabla_\theta J = \underbrace{\mathbb{E}\big[\nabla_\theta D\big]}_{\text{(i)}} + \underbrace{\mathbb{E}\big[D \cdot \nabla_\theta \log \pi_\theta(c \mid x)\big]}_{\text{(ii)}} + \underbrace{\mathbb{E}\big[D \cdot \nabla_\theta \log \pi_\theta(a \mid x, c)\big]}_{\text{(iii)}} \tag{14}$$

with all three expectations taken over $(c, a) \sim p_\theta$. This is exact; nothing has been dropped yet.

A.4 What "pathwise" means for term (i)

In $\nabla_\theta D(x, c, a)$, the token sequences c and a are held fixed as the concrete values that were sampled — there is no reparameterization through discrete tokens, and this gradient does not ask "how would different tokens have been sampled." It only asks: holding the realized trace and answer tokens fixed, how does the *forward pass* through those fixed tokens change as θ moves? Since $\pi_\theta(\cdot \mid x, c, a_{<t})$ is computed by running the transformer over the fixed sequence $(x, c, a_{<t})$, θ still appears via the weights that produce every hidden state and logit in that computation — including the key/value representations attended to from c. So term (i) is a genuine, fully differentiable gradient path; it just shapes *how the model represents and uses* a given (already-sampled) trace, not *which* trace tokens get sampled in the first place. Changing that requires reweighting the sampling probability of c itself — which is exactly what term (ii) does.

A.5 Why (iii) is dropped and (ii) is kept

This step is a modeling choice, not a further derivation — it is a bias/variance argument applied to the exact expression above.

- **Term (iii)** reweights the likelihood of the sampled *answer* by the same scalar D that already supervises the answer pointwise, position by position, via term (i). Because every answer position already receives a direct, per-position gradient target (the KL to p^* at that position), reweighting the whole answer's log-probability by the aggregate D on top of that is the standard "reward = per-step loss you can already differentiate through" case: it adds variance to the estimator without changing its expectation, since the pointwise objective is already being minimized pointwise. This is the same argument GKD/OPD use to discard the analogous on-policy credit-assignment term when dense per-token supervision is available.
- **Term (ii)** has no such redundancy. D depends on c only through its downstream effect on the answer distribution — there is no pointwise target for any individual trace token, so there is no pathwise gradient reaching trace-token logits from D at all (term (i)'s pathwise path shapes representations of a fixed c, but $\nabla_\theta \log \pi_\theta(c \mid x)$ is the only term whose gradient actually flows into the probability of choosing one trace token sequence over another). Dropping term (ii) therefore sets $\partial J/\partial$(trace-token logits) identically to zero — the reasoning policy would receive no learning signal whatsoever. Keeping it is what makes "dense supervision only on the answer" still able to shape the trace at all.

The resulting estimator actually used in training is Eq. (11) restricted to terms (i) and (ii), with D replaced by the group-baselined advantage A in term (ii) for variance reduction (§3) — the baseline subtraction is a standard control variate and does not change the expectation, since $\mathbb{E}_c[\nabla_\theta \log \pi_\theta(c \mid x)] = 0$.